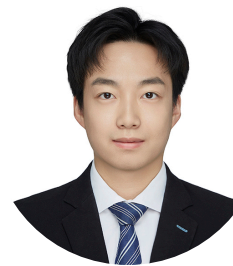


Wenhao Li

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📞 0009-0004-7752-6737 | 📧 Wenhao-Li-66 | 📧 PhD in Electronic Engineering

Ph.D. candidate in Information and Communication Engineering at Tsinghua University (Class of 2023), working on GNSS positioning and time-synchronization security with 3 first-author papers including IEEE Transactions on Smart Grid. Hands-on across embedded FPGA development, data science, and embodied intelligence, with a record of taking algorithms from theory to validated systems.



Education

 **Ph.D. in Information and Communication Engineering** 2023 - Present
TSINGHUA UNIVERSITY Beijing, China

- Research focus: GNSS positioning and time-synchronization security — 3 first-author papers published or accepted, including IEEE Transactions on Smart Grid
- Courses: Functional Analysis | Matrix Analysis and Applications | Pattern Recognition | Introduction to Artificial Intelligence

 **Bachelor of Engineering in Electronic Science and Technology** 2019 - 2023
TSINGHUA UNIVERSITY Beijing, China

- Thesis: Research and Implementation of GNSS Data Compression Technology for Power Estimation Applications
- Courses: Database | Computer Networks | Principles of Communications | Data and Algorithms | Signals and Systems
- GPA: 3.78/4.0

Publications

- W. Li**, M. Zhong, M. Lu, and H. Li, “Full-channel asynchronous spoofing technique for GNSS time synchronization attack,” *IEEE Trans. Smart Grid*, vol. 17, no. 2, pp. 1492–1502, Mar. 2026, doi: 10.1109/TSG.2025.3622566.
- W. Li**, M. Zhong, M. Lu, and H. Li, “Sequential separate clock drift anomaly detection method for GNSS time synchronization attack,” in *Proceedings of the 39th International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GNSS+ 2026)*, Orlando, Florida, Oct. 2026.
- W. Li**, L. Wang, M. Zhong, M. Lu, and H. Li, “An FD-DEFLATE data compression scheme for C/N0 estimation in GNSS interference monitoring,” in *Proceedings of the 37th International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GNSS+ 2024)*, Baltimore, Maryland, Oct. 2024, pp. 818–829. doi: 10.33012/2024.19871.
- M. Zhong, **W. Li**, M. Lu, and H. Li, “Feedback node aided distributed spoofing system for global navigation satellite system time synchronisation attack,” *IET Radar Sonar Navig.*, vol. 19, no. 1, p. e70088, 2025, doi: 10.1049/rsn2.70088.
- Y. Ma, H. Li, Z. Zhou, Z. Wu, **W. Li**, and M. Lu, “A fast C/N0 estimation method based on the ratio of acquisition correlation value,” in *China Satellite Navigation Conference (CSNC 2024) Proceedings*, C. Yang and J. Xie, Eds., Singapore: Springer Nature, 2024, pp. 567–577. doi: 10.1007/978-981-99-6932-6_47.

Professional Experience

 **Embodied Intelligence Algorithm Intern** 2025.06 - 2025.08
SERES GROUP Chongqing, China

- Targeted autonomous navigation for quadruped robots in unknown environments: surveyed VLM/VLN routes, then implemented a multimodal-perception navigation algorithm with a scene knowledge graph (SKG) that extended reliable path planning into cluttered, previously unseen scenes
- Diagnosed the root cause of parking-accuracy loss in the APA perception module by mapping how positioning error propagates through the control loop, then applied ESKF multi-sensor fusion with graph optimization to tighten pose estimation, improving parking accuracy and system robustness

Vision Language Navigation Automated Parking Assist

 **Algorithm R&D Intern** 2024.06 - 2024.08
NAVITECH Beijing, China

- Completed embedded development on a Xilinx FPGA platform in C, implementing and optimizing the hardware-software communication link to improve system response efficiency
- Designed a multi-peak signal capture module for unstable signal acquisition in complex scenarios, then ran robustness testing and performance evaluation that confirmed algorithm stability

Embedded Development Baseband Signal Processing

- Analyzed and mined large-scale datasets with SQL and Python, designing and validating regression prediction metrics that supported business decisions
 - Evaluated software functionality and practicality and optimized core performance indicators, improving overall system operating efficiency
- StatisticsData Processing

Projects

Vision-Language Navigation Application Research

SERES GROUP

2025.06 - 2025.08

Chongqing, China

- Addressed robot autonomy in unknown environments by researching VLM-based embodied navigation and building a scene knowledge graph (SKG) to strengthen path planning in complex scenes
 - Designed and implemented a multimodal feature-fusion strategy coupling visual perception with language understanding, enabling autonomous navigation and task execution for robots/agents without prior maps
- VLNVLMCVEmbodied AISKG

APA Positioning Error Propagation Research for Automated Parking System

SERES GROUP

2025.06 - 2025.08

Chongqing, China

- Investigated APA positioning accuracy by decomposing the error sources in the perception module and modeling how they propagate through the control system
 - Applied ESKF multi-sensor fusion with graph optimization to improve pose-estimation accuracy, giving parking path planning a more reliable positioning basis
- CVESKFGraph OptimizationAutonomous Driving

FPGA-based Streaming Data Compression Platform

PERSONAL PROJECT

2023.06 - 2024.04

Beijing, China

- Built an efficient streaming data compression architecture on an FPGA platform in Verilog
 - Designed a pipelined data-processing module that achieved low-latency, high-throughput compression of real-time data streams
- CompressionVerilogFPGADEFLATEZstandard

Subspace-based Face Recognition Technology

COURSE PROJECT

2024.05 - 2024.06

Beijing, China

- Studied subspace-based face representation and recognition, implementing dimensionality-reduction algorithms including SRC and Laplacian Eigenmaps
 - Constructed subspace models of face images to explore robust feature extraction and classification under varying illumination, expression, and occlusion
- SubspaceSRCFace IdentificationLaplacian

Audio-Video Joint Text Transcription Problem

COURSE PROJECT

2023.12 - 2024.01

Beijing, China

- Tackled joint recognition and transcription of audio-video bimodal signals by implementing multimodal information fusion
 - Extracted visual features with ResNet and modeled text sequences with BPE, improving speech recognition accuracy in complex scenarios
- BPEDLPythonResNet

Cocktail Party Effect Speaker Recognition Technology

COURSE PROJECT

2021.01 - 2021.02

Beijing, China

- Studied speaker recognition in complex acoustic environments by simulating target-speech separation and identification under the cocktail party effect
 - Used facial visual information to assist speech recognition, achieving accurate identity recognition in multi-speaker scenarios
- CVDLPythonSpeaker RecognitionFace Identification

Campus Experience

Deputy Secretary of the General Branch

DEPARTMENT OF ELECTRONIC ENGINEERING, TSINGHUA UNIVERSITY

2023.09 - 2024.06

Beijing, China

- Planned and coordinated external exchange activities for the Graduate Student League Committee and organized multiple department-level events, strengthening the committee's influence and cohesion

- Managed the full cycle of certification, graduation, and presentation activities for the Electronic Engineering Doctoral Lecturers Group, keeping operations efficient and orderly
- Connected with schools, universities, and enterprises inside and outside Tsinghua to expand presentation collaboration channels and raise the group’s social impact

- Organized and led the department’s practice team to Wuhan for in-depth field research and practical activities
- Won the department’s only gold award and placed in the top 4% university-wide (6/150), earning the “Excellent Team Leader” title

- Graded assignments and lab reports and ran regular Q&A sessions, helping the instructor refine teaching content and methods
- Compiled solution approaches and implementations for the Unscented Kalman Filter and Particle Filter, deepening students’ grasp of nonlinear non-Gaussian signal processing

- Graded assignments and tutored after class, explaining core algorithm principles to strengthen students’ application of statistical inference methods
- Co-authored exercises and case studies on the EM Algorithm, Kalman Filter, and Generalized Likelihood Ratio Test (GLRT), reinforcing the link between theory and practice

Certificates & Awards

2025	Tsinghua University Outstanding Teaching Assistant Award
2024	Tsinghua University Graduate Comprehensive Excellence Award
2024	Tsinghua University Outstanding Communist Youth League Member
2022	Tsinghua University Undergraduate Academic Excellence Award

Skills

Programming	Python MATLAB C C# Verilog SQL Java Rust
Positioning	GNSS RTK SLAM ESKF Graph Optimization
AI & Perception	VLN VLM CV Multimodal Perception SKG
Operating Systems	Linux Windows macOS
Languages	Chinese English Japanese